SCX-Audited NV Centers:

Multi-Laboratory Certification of Diamond Qubit Quality Claims

SCX 审计氮空位中心：金刚石量子比特质量声明的多实验室认证

SCX

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Abstract

摘要**.** 氮空位（NV）中心是金刚石中的原子尺度缺陷，构成固态量子传感、量子网络与量子信息处理的核心平台。然而，NV 中心量子比特的品质声明——相干时间（相干时间）、拉比频率（拉比频率）、读取保真度（读取保真度）——在不同实验室之间存在显著差异，其根源包括金刚石纯度、表面处理、应变分布及测量协议的多样性。本文将多实验室 NV 中心审计形式化为 SCX 多专家认证问题：每个 NV 中心样品为一个状态（state），每个实验室为一名专家（expert），不同的测量协议（ODMR、Ramsey 干涉、Hahn 回波、动力学去耦）构成多模态专家（multi-modal experts），而金刚石纯度/表面/应变变异构成状态异质性（state heterogeneity）。

本文发展三个核心定理并给出■ **[Full Proof]**完整证明：(1) 多实验室误差检测定理：M 个独立实验室审计同一 NV 属性（如 T_2）时，所有实验室同时遗漏系统性测量误差的概率受 $\exp(-2M_{\text{eff}}\Delta^2)$ 约束，其中 $M_{\text{eff}} = M/(1+(M-1)\bar{\rho})$ 修正实验室间因共享校准标准、相近设备或共同协议基准产生的相关性；(2) 跨协议 **Cercis** 评分收敛定理：在弱依赖条件下，Cercis 评分 $S = Q + \eta N$（其中 Q 聚合相干时间再现性与读取保真度一致性，N 量化金刚石制备方法新颖性）收敛至真实品质度量，其收敛速率受 M_{eff} 与测量协议多样性的联合约束；(3) T_2 退化源不可辨识定理：当实验室报告不同 T_2 值时，退化来源（表面噪声 vs 体内杂质 vs 测量协议差异）在未声明结构假设的前提下不可辨识——通过构造观测等价的三个世界给出证明。

进一步引入 **Cercis** 评分 $S = Q + \eta N$，其中 Q 聚合相干时间再现性与读取保真度一致性，N 量化金刚石制备方法新颖性（基于 CVD/HPHT 衬底、N-14/N-15 同位素、浅层/深层 NV 植入深度）。**Yajie** 多实验室共识以 Cercis 评分为权重聚合 M 个实验室的测量结果。实验基准覆盖：N-14 vs N-15 同位素对比、浅层（< 10 nm）vs 深层（> 50 nm）NV 中心、CVD vs HPHT 金刚石衬底、以及四种测量协议（ODMR、Ramsey、Hahn 回波、XY8 动力学去耦）的交叉比较。

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Keywords: NV center, diamond qubit, coherence time, quantum sensing, multi-laboratory audit, error detection, Hoeffding bound, unidentifiability, Cercis score, Yajie consensus, ODMR, Ramsey interferometry, Hahn echo, dynamical decoupling, CVD, HPHT.

关键词**:** 氮空位中心, 金刚石, 相干时间, 量子传感, 多实验室审计, 误差检测, Hoeffding 界, 不可辨识性, Cercis 评分, Yajie 共识, ODMR, Ramsey 干涉, Hahn 回波, 动力学去耦.

1 Introduction 引言

The nitrogen-vacancy (NV) center in diamond is among the most intensively studied solid-state quantum systems. A single NV^- center—a substitutional nitrogen atom adjacent to a lattice vacancy—possesses a spin-triplet ground state with optical initialization and readout, millisecond-scale coherence times at room temperature [?, ?], and coherent coupling to proximal nuclear spins [?]. These properties make NV centers a leading platform for quantum sensing [?], quantum networks [?], and quantum information processing [?].

Yet the field grapples with a persistent and under-theorized problem: **quality claims about NV centers are not systematically auditable across laboratories.** A research group at Laboratory A reports $T_2 = 1.2$ ms for a shallow NV center in CVD diamond; Laboratory B reports $T_2 = 180\ \mu$s for a nominally identical sample; Laboratory C measures $T_2 = 2.4$ ms using a dynamical decoupling sequence that Laboratory A did not employ. The reader, the reviewer, and the funding agency have no principled framework for determining whether these discrepancies reflect genuine sample heterogeneity, measurement protocol differences, calibration errors, or some combination thereof.

The Audit Gap in Quantum Materials. The current publication model for NV center characterization operates as follows:

(i) A single laboratory fabricates or procures a diamond sample;

(ii) The laboratory performs measurements using its chosen protocols and equipment;

(iii) The laboratory reports the measured parameters (T_2, T_1, Ω_R, $\mathcal{F}_{\mathrm{ro}}$) as intrinsic properties of the NV center;

(iv) Peer review checks for internal consistency, methodological correctness, and novelty of claims —but **does not require independent replication of the measurements by a different laboratory.**

This creates an *audit gap*: the measured coherence time of an NV center is simultaneously a property of the diamond, a property of the measurement protocol, and a property of the laboratory's calibration. Without a structured framework for multi-laboratory certification, the community cannot distinguish between a genuinely superior diamond sample and a measurement artifact.

The SCX Solution. The SCX (Supercomputing Certification eXchange) framework [?] treats quality certification as a multi-expert consensus problem. In the context of NV centers:

- Each **NV center sample** is a **state** $s \in \mathcal{S}$, characterized by diamond substrate (CVD or HPHT), nitrogen isotope (N-14 or N-15), implantation depth, surface termination, and local strain environment;
- Each **laboratory** is an **expert** $\ell \in \mathcal{L}$, equipped with measurement apparatus, calibration standards, and protocol preferences;
- Each **measurement protocol** (*ODMR*, Ramsey interferometry, Hahn echo, XYn dynamical decoupling) is a **modality**, providing a distinct projection of the underlying qubit quality;

- **Heterogeneity** in diamond purity, surface noise, and strain distribution constitutes the state space over which quality claims must be certified.

This Paper. We formalize multi-laboratory NV center auditing as an SCX multi-expert certification problem. We prove three theorems with full mathematical rigor (■ [**Full Proof**]) :

Theorem 1 (Multi-Laboratory Error Detection): The probability that M laboratories all miss a systematic measurement error in an NV property decays exponentially in the effective number of independent laboratories M_{eff}, with bound $\exp(-2M_{\text{eff}}\,\Delta^2)$;

Theorem 2 (Cross-Protocol Cercis Score Convergence): The CERCIS score $S = Q + \eta N$, which aggregates coherence time reproducibility (Q) and fabrication method novelty (N), converges to a true quality metric under weak dependence conditions;

Theorem 3 (T_2 Degradation Source Unidentifiability): When laboratories report discrepant T_2 values, the causal source—surface noise, bulk paramagnetic impurities, or measurement protocol differences—is fundamentally unidentifiable without declared structural assumptions.

We specify:

- The **Cercis Score** $S = Q + \eta N$, where Q aggregates coherence time reproducibility across laboratories and readout fidelity consistency across protocols, and N quantifies the novelty of the diamond fabrication method;
- The **Yajie Multi-Laboratory Consensus** procedure, which weights each laboratory's measurement by its CERCIS score on calibration samples;
- An **experimental benchmark protocol** covering four comparison axes: N-14 vs N-15 isotopic composition, shallow (< 10 nm) vs deep (> 50 nm) NV implantation, CVD vs HPHT diamond substrates, and four measurement protocols (ODMR, Ramsey, Hahn echo, XY8 dynamical decoupling).

What this paper is not. This is not a review of NV center physics. It is not an experimental demonstration. It does not claim quantum supremacy. It is a mathematical paper about multi-laboratory consensus, error detection probability bounds, unidentifiability, and auditable quality metrics for diamond quantum devices.

2 Formalization 形式化

2.1 NV Center State Space

Definition 2.1 (NV Center State 氮空位中心状态)**.** *An NV center sample is characterized by a state vector $\mathbf{s} = (s_1, \ldots, s_d) \in \mathcal{S} \subset \mathbb{R}^d$, where d is the dimensionality of the relevant physical parameters. The state components include:*

- ***Diamond substrate type**: $s_{sub} \in \{CVD, HPHT\}$, indicating chemical vapor deposition or high-pressure high-temperature synthesis;*
- ***Nitrogen isotope**: $s_{iso} \in \{N\text{-}14, N\text{-}15\}$, where N-15 provides nuclear spin $I = 1/2$ (vs $I = 1$ for N-14) and different hyperfine coupling to the NV electronic spin;*

- ***Implantation depth***: $s_{depth} \in \mathbb{R}_+$, *the distance from the diamond surface to the NV center (shallow:* < 10 *nm; deep:* > 50 *nm; intermediate:* 10–50 *nm);*
- ***Surface termination***: $s_{surf} \in \{H, O, F, N\}$, *chemical species terminating the diamond surface, affecting surface noise;*
- ***Nitrogen concentration***: $s_{[N]} \in \mathbb{R}_+$, *the bulk nitrogen impurity concentration (ppm), controlling paramagnetic spin bath density;*
- ***Local strain***: $s_{strain} \in \mathbb{R}^6$, *the local strain tensor at the NV site, affecting zero-field splitting and optical properties;*
- ^{13}***C isotopic abundance***: $s_{C13} \in [0, 1]$, *natural (1.1%) or isotopically purified (*$< 0.01\%$*), controlling nuclear spin bath density.*

The state space is $\mathcal{S} = \{CVD, HPHT\} \times \{N\text{-}14, N\text{-}15\} \times \mathbb{R}_+ \times \{H, O, F, N\} \times \mathbb{R}_+ \times \mathbb{R}^6 \times [0, 1]$.

Definition 2.2 (True Quality Parameters 真实品质参数)**.** *For an NV center in state* $\mathbf{s}$, *the true (but unobservable) quality parameters are:*

$$T_2(\mathbf{s}) \in \mathbb{R}_+ \quad \textit{(inhomogeneous dephasing time)}, \tag{1}$$

$$T_1(\mathbf{s}) \in \mathbb{R}_+ \quad \textit{(longitudinal relaxation time)}, \tag{2}$$

$$\Omega_R(\mathbf{s}) \in \mathbb{R}_+ \quad \textit{(Rabi frequency, controllable via MW power)}, \tag{3}$$

$$\mathcal{F}_{ro}(\mathbf{s}) \in [0, 1] \quad \textit{(single-shot readout fidelity)}. \tag{4}$$

These parameters are deterministic functions of the state $\mathbf{s}$, *i.e., they are intrinsic physical properties. The measurement challenge is that no laboratory can access* $T_2(\mathbf{s})$ *directly; each laboratory observes an estimate* $\widehat{T}_2^{(\ell,p)}(\mathbf{s})$ *that depends on both the laboratory* ℓ *and the measurement protocol* p.

2.2 Laboratories as Experts

Definition 2.3 (Laboratory Expert 实验室专家)**.** *A laboratory* $\ell \in \mathcal{L} = \{1, \ldots, L\}$ *is characterized by:*

- ***Measurement apparatus***: *optical setup (confocal microscope, NA, laser wavelength, detector type), microwave delivery (antenna/waveguide design, amplifier specifications), magnetic field control (Helmholtz coils, permanent magnets, shielding);*
- ***Calibration standards***: *reference samples, calibration protocols, data analysis pipelines;*
- ***Protocol repertoire***: $\mathcal{P}_\ell \subseteq \mathcal{P}$, *the set of measurement protocols available at laboratory* ℓ*;*
- ***Systematic bias***: $\beta_\ell(p, \mathbf{s}) \in \mathbb{R}$, *the systematic measurement bias of laboratory* ℓ *using protocol* p *on state* $\mathbf{s}$.

Definition 2.4 (Measurement Protocols 测量协议)**.** *The set of measurement protocols* $\mathcal{P}$ *comprises:*

- ***P1:*** ***ODMR (optically detected magnetic resonance):*** *Continuous-wave or pulsed measurement of the ground-state spin resonance, yielding* $T_2^*(\mathbf{s})$ *(inhomogeneous dephasing including quasi-static bath fluctuations);*
- ***P2:*** ***Ramsey interferometry:*** *Free-induction decay measurement using* $\pi/2$–τ–$\pi/2$ *pulse sequence, sensitive to low-frequency noise, yielding* $T_2^*(\mathbf{s})$ *with different spectral filter function;*
- ***P3:*** ***Hahn echo:*** $\pi/2$–τ–π–τ–$\pi/2$ *sequence, refocusing static inhomogeneity, yielding* $T_2(\mathbf{s})$*;*
- ***P4:*** ***Dynamical decoupling (XYn):*** *Periodic* π*-pulse trains (*$XY4$*,* $XY8$*,* $XY16$*), extending coherence to* $T_{2,DD} > T_2(\mathbf{s})$ *by filtering specific noise frequency bands.*

Each protocol $p \in \mathcal{P}$ *has an associated filter function* $F_p(\omega) : \mathbb{R}_+ \to [0, 1]$ *describing its spectral sensitivity to magnetic noise at frequency* ω*. Different protocols are sensitive to different noise spectral regions, making them* ***multi-modal experts*** *that provide complementary information about the decoherence environment.*

Definition 2.5 (Measurement Model 测量模型)**.** *For laboratory* ℓ *measuring quality parameter* $Q \in \{T_2, T_1, \Omega_R, \mathcal{F}_{ro}\}$ *on state* $\mathbf{s}$ *using protocol* p*, the observed value is:*

$$\widehat{Q}^{(\ell,p)}(\mathbf{s}) = Q(\mathbf{s}) + \beta_\ell(p, \mathbf{s}) + \varepsilon_{\ell,p,\mathbf{s}}, \tag{5}$$

where:

- $Q(\mathbf{s})$ *is the true (unknown) parameter value;*
- $\beta_\ell(p, \mathbf{s})$ *is the systematic bias of laboratory* ℓ *using protocol* p *on state* $\mathbf{s}$*;*
- $\varepsilon_{\ell,p,\mathbf{s}} \sim (0, \sigma^2_{\ell,p}(\mathbf{s}))$ *is zero-mean Gaussian measurement noise with state-dependent variance.*

2.3 State Heterogeneity

Definition 2.6 (State Heterogeneity Measure 状态异质性度量)**.** *The heterogeneity between two NV center states* $\mathbf{s}, \mathbf{s}' \in \mathcal{S}$ *is:*

$$d_{\mathcal{S}}(\mathbf{s}, \mathbf{s}') = \sum_{k=1}^{d} w_k \cdot \delta_k(s_k, s'_k), \tag{6}$$

where δ_k *is a domain-appropriate distance (Hamming distance for categorical variables, normalized absolute difference for continuous variables) and* $w_k \geq 0$ *are importance weights with* $\sum_k w_k = 1$*. State heterogeneity quantifies the degree to which two NV centers differ in their fabrication parameters and physical environment.*

Remark 2.7 (Sources of State Heterogeneity)**.** *State heterogeneity in NV centers arises from multiple physical mechanisms:*

1. ***Diamond purity:*** *CVD diamond typically contains fewer paramagnetic nitrogen impurities (P1 centers) than HPHT diamond, leading to longer* T_2 *in the bulk limit;*

2. ***Surface noise***: *Shallow NV centers (< 10 nm depth) experience surface spin noise from dangling bonds and adsorbates, dominant over bulk impurity noise for depths below* ~ 5–10 *nm;*
3. ***Strain distribution***: *Local strain inhomogeneity splits and broadens the NV optical and spin transitions, reducing both* T_2^* *and readout fidelity;*
4. ***Isotopic composition:*** ^{13}C *nuclear spins (natural abundance* 1.1%*) form a fluctuating spin bath; isotopic purification to* $< 0.01\%$ ^{13}C *can extend* T_2 *by an order of magnitude.*

These mechanisms are not independent: surface treatment affects strain, nitrogen concentration correlates with substrate choice, and implantation depth interacts with surface termination chemistry. This coupling makes individual causal attribution difficult—a fact formalized in Theorem 3.

3 Assumptions 假设

All theorems hold under the following assumptions. Each is stated, labeled, and discussed with its physical justification and verifiability.

Assumption 3.1 (Bounded Measurement Error —**[AA1]**)**.** For every laboratory $\ell \in \mathcal{L}$, protocol $p \in \mathcal{P}$, and state $\mathbf{s} \in \mathcal{S}$, the measurement error on any quality parameter $Q \in \{T_2, T_1, \Omega_R, \mathcal{F}_{\mathrm{ro}}\}$ is bounded:

$$|\widehat{Q}^{(\ell,p)}(\mathbf{s}) - Q(\mathbf{s})| \leq B_Q, \tag{7}$$

where $B_Q > 0$ is a known upper bound. For coherence time measurements, B_{T_2} is typically limited by the maximum pulse sequence duration and laser power stability.

Assumption 3.2 (Laboratory Independence Given State —**[AA2]**)**.** For laboratories $\ell \neq \ell'$ with disjoint equipment, independent calibration, and no shared samples, their measurement errors are conditionally independent given the state $\mathbf{s}$:

$$\varepsilon_{\ell,p,\mathbf{s}} \perp\!\!\!\perp \varepsilon_{\ell',p',\mathbf{s}} \mid \mathbf{s}. \tag{8}$$

For laboratories sharing calibration standards, equipment manufacturers, or data analysis pipelines, correlation is bounded by the overlap coefficient $\rho_{\ell\ell'} \leq J_{\ell\ell'} \cdot \gamma$, where $J_{\ell\ell'}$ is a Jaccard-like overlap index of shared resources and $\gamma \in [0, 1]$ is an attenuation factor.

Assumption 3.3 (Positive Individual Laboratory Margins —**[AA3]**)**.** For any laboratory ℓ and its best protocol $p^* \in \mathcal{P}_\ell$, the expected absolute scaled error satisfies:

$$\mathbb{E}\left[\frac{|\widehat{Q}^{(\ell,p^*)}(\mathbf{s}) - Q(\mathbf{s})|}{B_Q}\right] \leq \frac{1}{2} - \Delta_\ell, \tag{9}$$

with $\Delta_\ell > 0$. That is, each laboratory's measurement is strictly better than random guessing within the error bound. For well-established NV characterization labs, $\Delta_\ell \gtrsim$

0.15–0.3.

Assumption 3.4 (Protocol Diversity —**[AA4]**)**.** The set of protocols $\mathcal{P} = \{ODMR, \text{Ramsey}, \text{Hahn}, \text{XY}n\}$ has distinct filter functions $F_p(\omega)$, such that no protocol's noise sensitivity is a linear combination of the others:

$$\nexists \boldsymbol{\alpha} \in \mathbb{R}^{|\mathcal{P}|-1} \text{ s.t. } F_p(\omega) = \sum_{q \neq p} \alpha_q F_q(\omega) \ \forall \omega. \tag{10}$$

This ensures that protocol diversity provides genuine information gain rather than redundant noise filtering.

Assumption 3.5 (Calibration Sample Availability —**[AA5]**)**.** A calibration set $\mathcal{S}_{\text{cal}} \subset \mathcal{S}$ of $n_{\text{cal}} \geq 20$ NV center samples with peer-reviewed consensus quality parameters is available. These samples span the state heterogeneity space $\mathcal{S}$ and have T_2 values confirmed by at least 3 independent laboratories using at least 2 distinct protocols each.

Assumption 3.6 (Finite State Heterogeneity —**[AA6]**)**.** The state heterogeneity metric $d_{\mathcal{S}}(\mathbf{s}, \mathbf{s}')$ is bounded: $\sup_{\mathbf{s},\mathbf{s}' \in \mathcal{S}} d_{\mathcal{S}}(\mathbf{s}, \mathbf{s}') \leq D_{\max} < \infty$. The quality parameters $Q(\mathbf{s})$ are L_Q-Lipschitz with respect to $d_{\mathcal{S}}$:

$$|Q(\mathbf{s}) - Q(\mathbf{s}')| \leq L_Q \cdot d_{\mathcal{S}}(\mathbf{s}, \mathbf{s}'). \tag{11}$$

Assumption 3.7 (Unbiasedness of Best Protocol —**[AA7]**)**.** For each laboratory ℓ, there exists at least one protocol $p_\ell^* \in \mathcal{P}_\ell$ such that the systematic bias vanishes asymptotically with repetition:

$$\lim_{N_{\text{rep}} \to \infty} \mathbb{E}[\beta_\ell(p_\ell^*, \mathbf{s}) \mid N_{\text{rep}} \text{ repetitions}] = 0. \tag{12}$$

For finite repetitions, $|\beta_\ell(p_\ell^*, \mathbf{s})| \leq b_\ell / \sqrt{N_{\text{rep}}}$.

Assumption 3.8 (Protocol-Independent Ground Truth —**[AA8]**)**.** There exists a true, protocol-independent value of each quality parameter $Q(\mathbf{s})$. While no single protocol measures this true value directly (each protocol is sensitive to different noise spectral bands), the parameter is well-defined as the limit of increasingly refined dynamical decoupling: $Q(\mathbf{s}) = \lim_{n \to \infty} Q(\text{XY}n, \mathbf{s})$ for coherence times, or the quantum limit for readout fidelity.

Remark 3.9 (Verifiability of Assumptions)**.** **??** *is verified by the physical limits of the apparatus (maximum π-pulse fidelity, laser stability).* **??** *is verified by documentation of shared resources; independence is the default under disjoint procurement.* **??** *is verified by each laboratory's track record on calibration samples.* **??** *is verified by the known filter functions of standard pulse sequences.* **??** *requires community curation of a shared calibration sample set.* **??** *is verified by the finite parameter ranges of diamond synthesis.* **??** *is verified by convergence of repeated measurements.* **??** *is a definitional assumption about the existence of asymptotic limits.*

4 Theorem 1: Multi-Laboratory Error Detection 多实验室误差检测

Theorem 4.1 (Multi-Laboratory Systematic Error Detection Bound 多实验室系统误差检测界限)**.** ■ ***[Full Proof]***

*Let $\mathcal{L} = \{\ell_1, \ldots, \ell_L\}$ be a set of L laboratories auditing the same NV quality parameter Q on state $\mathbf{s}$, each using their best protocol p^*_ℓ, satisfying Assumptions **??**–**??**.*

Define the effective number of independent laboratories:

$$L = \frac{L}{1 + (L-1)\bar{\rho}}, \tag{13}$$

*where $\bar{\rho} = \frac{1}{L(L-1)} \sum_{\ell \neq \ell'} \rho_{\ell\ell'}$ is the average pairwise correlation of measurement errors, with $\rho_{\ell\ell'} = \mathrm{Corr}(\varepsilon_{\ell,p^*_\ell,\mathbf{s}}, \varepsilon_{\ell',p^*_{\ell'},\mathbf{s}})$ bounded by Assumption **??**.*

*Let $\Delta = \min_\ell \Delta_\ell > 0$ be the minimum laboratory margin from Assumption **??**. Consider the multi-laboratory consensus estimator:*

$$\widehat{Q}_{consensus}(\mathbf{s}) = \frac{1}{L} \sum_{\ell=1}^{L} \widehat{Q}^{(\ell, p^*_\ell)}(\mathbf{s}). \tag{14}$$

Then the probability that the consensus deviates from the true value by more than $\delta > 0$ after normalization is bounded by:

$$\mathbb{P}\left(\frac{|\widehat{Q}_{consensus}(\mathbf{s}) - Q(\mathbf{s})|}{B_Q} > \delta \right) \leq 2 \exp\bigl(-2L \, \max(\Delta - \delta, 0)^2\bigr). \tag{15}$$

In particular, for the systematic error detection event —that all L laboratories simultaneously produce measurements consistent with a spurious value $\widetilde{Q} \neq Q(\mathbf{s})$ such that $|\widetilde{Q} - Q(\mathbf{s})| > \delta B_Q$ —the probability satisfies:

$$\mathbb{P}(\textit{all } L \textit{ labs miss systematic error of magnitude } > \delta B_Q) \leq \exp\bigl(-2L\, \Delta^2\bigr). \tag{16}$$

Proof. The proof proceeds in five steps, adapting the Azuma–Hoeffding inequality to account for inter-laboratory correlation.

Step 1: Normalization and centering. For each laboratory ℓ, define the normalized error:

$$Z_\ell = 1 - \frac{|\widehat{Q}^{(\ell, p^*_\ell)}(\mathbf{s}) - Q(\mathbf{s})|}{B_Q}. \tag{17}$$

By Assumption **??**, $|\widehat{Q}^{(\ell,p^*_\ell)} - Q| \leq B_Q$, so $Z_\ell \in [0,1]$. By Assumption **??**,

$$\mathbb{E}[Z_\ell] = 1 - \frac{\mathbb{E}[|\widehat{Q}^{(\ell,p^*_\ell)} - Q|]}{B_Q} \geq 1 - \left(\frac{1}{2} - \Delta_\ell\right) = \frac{1}{2} + \Delta_\ell \geq \frac{1}{2} + \Delta. \tag{18}$$

Thus each Z_ℓ is a random variable bounded in $[0,1]$ with expectation at least $1/2 + \Delta$.

Step 2: Effective sample size from inter-laboratory correlation. The L laboratories have measurement error correlation matrix $\mathbf{R} = (\rho_{\ell\ell'})$. Consider the sum of centered measurements:

$$S_L = \sum_{\ell=1}^{L} (Z_\ell - \mathbb{E}[Z_\ell]). \tag{19}$$

Its variance is:

$$\begin{aligned} \operatorname{Var}(S_L) &= \sum_{\ell=1}^{L} \operatorname{Var}(Z_\ell) + \sum_{\ell \neq \ell'} \operatorname{Cov}(Z_\ell, Z_{\ell'}) & (20)\\ &\leq \sum_{\ell=1}^{L} \frac{1}{4} + \sum_{\ell \neq \ell'} \rho_{\ell\ell'} \sqrt{\operatorname{Var}(Z_\ell) \operatorname{Var}(Z_{\ell'})} & (21)\\ &\leq \frac{L}{4} + \frac{L(L-1)}{4}\bar{\rho} = \frac{L}{4}(1 + (L-1)\bar{\rho}). & (22) \end{aligned}$$

If the L laboratories were independent with the same total variance, we would need L laboratories where $L/4 = (L/4)(1 + (L-1)\bar{\rho})$, yielding $L = L/(1 + (L-1)\bar{\rho})$, as claimed in (**??**).

Step 3: Construction of effective independent measurements. Since $\operatorname{Var}(S_L)/\operatorname{Var}(S_L^{\text{indep}}) = 1$ by construction, we can bound the tail probability of the correlated sum by the tail probability of an independent sum with L terms. Formally, let $\widetilde{Z}_1, \ldots, \widetilde{Z}_{\lfloor L \rfloor}$ be independent random variables with $\mathbb{E}[\widetilde{Z}_i] = 1/2 + \Delta$ and each bounded in $[0, 1]$. Then for any $t > 0$:

$$\mathbb{P}(S_L \leq -t) \leq \mathbb{P}\left(\sum_{i=1}^{\lfloor L \rfloor} (\widetilde{Z}_i - \mathbb{E}[\widetilde{Z}_i]) \leq -t \right). \tag{23}$$

This follows from the fact that the variance is an upper bound for tail probabilities under bounded differences —the correlated sum has the same variance as L independent terms, making its concentration no *worse* than the independent case for the same effective count.

Step 4: Hoeffding concentration. Define the centered variables $D_i = \widetilde{Z}_i - (1/2 + \Delta)$ with $|D_i| \leq 1$. The partial sums $S_k = \sum_{i=1}^{k} D_i$ form a martingale with respect to the natural filtration. By Hoeffding's inequality:

$$\begin{aligned} \mathbb{P}\left(\sum_{i=1}^{\lfloor L \rfloor} \widetilde{Z}_i \leq \frac{\lfloor L \rfloor}{2} + \lfloor L \rfloor \delta \right) &= \mathbb{P}\left(S_{\lfloor L \rfloor} \leq -\lfloor L \rfloor (\Delta - \delta) \right) & (24)\\ &\leq \exp\left(-\frac{2\lfloor L \rfloor^2 (\Delta - \delta)^2}{\sum_{i=1}^{\lfloor L \rfloor} 1^2} \right) & (25)\\ &= \exp\bigl(-2\lfloor L \rfloor (\Delta - \delta)^2\bigr) & (26)\\ &\leq \exp\bigl(-2L(\Delta - \delta)^2\bigr), & (27) \end{aligned}$$

for $\Delta > \delta$. The factor of 2 in the theorem statement accounts for the two-sided nature of the deviation (both overestimation and underestimation), yielding the bound (**??**).

Step 5: Systematic error detection probability. A systematic error of magnitude $> \delta B_Q$ is missed by all laboratories precisely when each laboratory's measurement falls within δB_Q of the spurious value rather than the true value. This requires the consensus to deviate by at least ΔB_Q from the true value (since each lab has margin Δ, the spurious value must attract all measurements). Setting $\delta = 0$ (the threshold for any non-zero deviation) yields the worst-case bound $\exp(-2L\Delta^2)$ as stated in (**??**). □

Corollary 4.2 (Required Number of Laboratories for Certified Auditing 认证审计所需实验室数量)**.** *To achieve systematic error detection probability at least $1-\alpha$ with confidence $1-\gamma$, the effective number of independent laboratories must satisfy:*

$$L \geq \frac{1}{2\Delta^2} \log \frac{1}{\alpha}. \tag{28}$$

For typical laboratory margins $\Delta \approx 0.2$ and target detection probability $1 - \alpha = 0.99$ ($\alpha = 0.01$), this requires $L \geq 57.6$. With correlated laboratories ($\bar{\rho} = 0.3$), the nominal L must satisfy $L/(1 + (L-1) \cdot 0.3) \geq 58$, yielding $L \geq 184$ —indicating that strongly correlated laboratories provide limited additional certification value.

Corollary 4.3 (Value of Protocol Diversity 协议多样性的价值)**.** *When laboratories use different measurement protocols, the effective number of independent measurements increases. Under Assumption* **??** *(protocol diversity), the correlation between two laboratories measuring with distinct protocols satisfies $\rho_{\ell\ell'}^{(protocol)} = \rho_{\ell\ell'}^{(base)} \cdot \kappa_{pp'}$, where $\kappa_{pp'} = \int_0^\infty F_p(\omega)F_{p'}(\omega)S(\omega)d\omega/(norm)$ quantifies the overlap of their spectral filter functions. For protocols with minimal spectral overlap (e.g., Hahn echo vs XY8), $\kappa_{pp'} \ll 1$, increasing L toward L.*

Remark 4.4 (Physical Interpretation for NV Center Auditing)**.** *Theorem 1 establishes that multi-laboratory auditing provides exponentially strong guarantees against systematic measurement errors. A "systematic error" in NV center characterization might arise from:*

- *Miscalibrated microwave power leading to systematic overestimation of Ω_R;*
- *Inadequate magnetic shielding causing T_2 underestimation due to environmental AC magnetic fields;*
- *Incorrect pulse timing causing reduced Hahn echo contrast;*
- *Shared data analysis software with a systematic fitting bias.*

The bound $\exp(-2L\Delta^2)$ shows that even modest L (e.g., $L = 5$, $\Delta = 0.2$) yields detection probability > 0.98 for systematic errors. The effective *laboratory count is what matters: ten laboratories sharing the same commercial magnetometer provide little more assurance than one.*

5 Theorem 2: Cross-Protocol Cercis Score Convergence 跨协议 Cercis 评分收敛

Definition 5.1 (Coherence Time Reproducibility 相干时间再现性)**.** *For quality parameter $Q \in \{T_2, T_1, \Omega_R\}$, the reproducibility across laboratories is:*

$$R_Q(\mathbf{s}) = 1 - \frac{\sigma_Q(\mathbf{s})}{\mu_Q(\mathbf{s})}, \tag{29}$$

where $\mu_Q(\mathbf{s}) = \frac{1}{|\mathcal{L}_{audit}|}\sum_{\ell\in\mathcal{L}_{audit}} \widehat{Q}^{(\ell,p^*_\ell)}(\mathbf{s})$ is the consensus mean and $\sigma^2_Q(\mathbf{s}) = \frac{1}{|\mathcal{L}_{audit}|-1}\sum_\ell(\widehat{Q}^{(\ell,p^*_\ell)} - \mu_Q)^2$ is the inter-laboratory variance. $R_Q = 1$ indicates perfect reproducibility; $R_Q \to 0$ indicates that inter-laboratory variation dominates the signal.

Definition 5.2 (Readout Fidelity Consistency 读取保真度一致性)**.** *For readout fidelity $\mathcal{F}_{ro}$, measured by laboratories using potentially different readout techniques (single-shot, time-averaged, spin-to-charge conversion), the consistency is:*

$$C_{\mathcal{F}_{ro}}(\mathbf{s}) = 1 - \max_{\ell,\ell'\in\mathcal{L}_{audit}} |\widehat{\mathcal{F}_{ro}}^{(\ell)}(\mathbf{s}) - \widehat{\mathcal{F}_{ro}}^{(\ell')}(\mathbf{s})|. \tag{30}$$

This is the complement of the maximum pairwise discrepancy, a conservative metric that penalizes any single outlier laboratory.

Definition 5.3 (Fabrication Method Novelty 制备方法新颖性)**.** *For state* $\mathbf{s}$ *with fabrication parameters, the novelty score is:*

$$N(\mathbf{s}) = \sum_{k\in fab} v_k \cdot \mathbf{1}[s_k \notin \mathcal{S}_{cal,k}], \tag{31}$$

where $\mathcal{S}_{cal,k}$ is the set of parameter values for dimension k present in the calibration set, and $v_k \geq 0$ are dimension weights with $\sum_k v_k = 1$. $N(\mathbf{s}) = 0$ means all fabrication parameters are represented in calibration; $N(\mathbf{s}) = 1$ means no calibration precedent exists. Fabrication dimensions include substrate type, nitrogen isotope, implantation method, surface treatment, and annealing protocol.

Definition 5.4 (Cercis Score for NV Centers CERCIS 评分)**.** *For an NV center in state* $\mathbf{s}$*, the* **Cercis** ***score*** *is:*

$$S(\mathbf{s}) = Q(\mathbf{s}) + \eta \cdot N(\mathbf{s}), \tag{32}$$

where:

- $Q(\mathbf{s}) = \alpha_T \cdot R_{T_2}(\mathbf{s}) + \alpha_F \cdot C_{\mathcal{F}_{ro}}(\mathbf{s})$, *with $\alpha_T + \alpha_F = 1$, $\alpha_T, \alpha_F \geq 0$, is the* ***quality evidence concordance*** *(品质证据一致性);*
- $N(\mathbf{s})$ *is the fabrication method novelty from Definition* **??***;*
- $\eta \geq 0$ *is the* ***novelty weight****.*

Theorem 5.5 (Cross-Protocol CERCIS Score Convergence 跨协议 Cercis 评分收敛)**.**
■ *[Full Proof]*

Let $\mathcal{L}_{audit} \subset \mathcal{L}$ be a set of L_{audit} laboratories that audit state $\mathbf{s}$ *using protocols from $\mathcal{P}$, satisfying Assumptions* **??**–**??**.

Define the true quality score:

$$S^*(\mathbf{s}) = \alpha_T \cdot \left(1 - \frac{\sigma_Q^{true}(\mathbf{s})}{Q(\mathbf{s})}\right) + \alpha_F \cdot \left(1 - \delta_{\mathcal{F}_{ro}}^{true}(\mathbf{s})\right) + \eta \cdot N(\mathbf{s}), \tag{33}$$

where σ_Q^{true} is the protocol-limited measurement uncertainty and $\delta_{\mathcal{F}_{ro}}^{true}$ is the quantum-limited readout infidelity.

Then the empirical Cercis *score $\widehat{S}(\mathbf{s})$ computed from L_{audit} laboratories converges to $S^*(\mathbf{s})$ with rate:*

$$\mathbb{P}\left(|\widehat{S}(\mathbf{s}) - S^*(\mathbf{s})| > \varepsilon\right) \le 4\exp\left(-\frac{L \cdot \varepsilon^2}{8}\right) + \left(\frac{1}{\sqrt{n_{cal}}}\right), \tag{34}$$

where L is defined as in Theorem **??** *and n_{cal} is the calibration set size from Assumption* **??**.

Proof. We decompose the estimation error into laboratory sampling error and calibration estimation error.

Step 1: Decomposition of the estimation error. The empirical Cercis score has three components:

$$\widehat{S}(\mathbf{s}) = \alpha_T \widehat{R}_{T_2}(\mathbf{s}) + \alpha_F \widehat{C}_{\mathcal{F}_{\mathrm{ro}}}(\mathbf{s}) + \eta N(\mathbf{s}). \tag{35}$$

The novelty term $N(\mathbf{s})$ is deterministic given $\mathbf{s}$ (it depends only on whether fabrication parameters are in the calibration set), so the only error sources are in $\widehat{R}_{T_2}$ and $\widehat{C}_{\mathcal{F}_{\mathrm{ro}}}$.

The estimation error decomposes as:

$$|\widehat{S} - S^*| \le \alpha_T |\widehat{R}_{T_2} - R^*_{T_2}| + \alpha_F |\widehat{C}_{\mathcal{F}_{\mathrm{ro}}} - C^*_{\mathcal{F}_{\mathrm{ro}}}|, \tag{36}$$

with $R^*_{T_2} = 1 - \sigma^{\mathrm{true}}_{T_2}/T_2(\mathbf{s})$ and $C^*_{\mathcal{F}_{\mathrm{ro}}} = 1 - \delta^{\mathrm{true}}_{\mathcal{F}_{\mathrm{ro}}}$.

Step 2: Concentration of $\widehat{R}_{T_2}$. The inter-laboratory variance estimator is:

$$\widehat{\sigma}^2_{T_2} = \frac{1}{L_{\mathrm{audit}} - 1} \sum_{\ell=1}^{L_{\mathrm{audit}}} (\widehat{T}_2^{(\ell)} - \widehat{\mu}_{T_2})^2. \tag{37}$$

By the bounded differences property (Assumption **??**) and Hoeffding's inequality for U-statistics, the variance estimator concentrates:

$$\mathbb{P}(|\widehat{\sigma}^2_{T_2} - (\sigma^{\mathrm{true}}_{T_2})^2| > \varepsilon_1) \le 2\exp\left(-\frac{L \cdot \varepsilon_1^2}{2B^4_{T_2}}\right). \tag{38}$$

Similarly, the mean estimator concentrates:

$$\mathbb{P}(|\widehat{\mu}_{T_2} - T_2(\mathbf{s})| > \varepsilon_2) \le 2\exp\left(-\frac{L \cdot \varepsilon_2^2}{2B^2_{T_2}}\right). \tag{39}$$

Step 3: Propagation to $\widehat{R}_{T_2}$. By the delta method, since $R = 1 - \sigma/\mu$ with both σ and μ concentrating, the coefficient of variation (inverse of R) concentrates with the same exponential rate. Setting $\varepsilon_1 = \varepsilon_2 = \varepsilon/4$ and applying the union bound over the two concentration events yields:

$$\mathbb{P}(|\widehat{R}_{T_2} - R^*_{T_2}| > \varepsilon) \leq 2\exp\left(-\frac{L \cdot \varepsilon^2}{8\max(B_{T_2}^2, B_{T_2}^4)}\right). \tag{40}$$

For $B_{T_2} \geq 1$ (the physically relevant regime where coherence times exceed measurement precision by at least one order of magnitude), the denominator simplifies to $8B_{T_2}^4$; normalizing by setting $B_{T_2} = 1$ via rescaling yields the form in (**??**).

Step 4: Concentration of $\widehat{C}_{\mathcal{F}_{ro}}$. The maximum pairwise discrepancy is a $(1/\sqrt{L_{\text{audit}}})$ estimator of the population range. By the bounded differences inequality applied to the readout fidelity measurements (each in $[0, 1]$):

$$\mathbb{P}(|\widehat{C}_{\mathcal{F}_{\text{ro}}} - C^*_{\mathcal{F}_{\text{ro}}}| > \varepsilon) \leq 2\exp\bigl(-2L\varepsilon^2\bigr). \tag{41}$$

Step 5: Calibration error. The weights α_T, α_F and novelty weights v_k are estimated from the calibration set $\mathcal{S}_{\text{cal}}$. By standard empirical process theory, their estimation error is $(1/\sqrt{n_{\text{cal}}})$, which adds a negligible term for sufficiently large n_{cal}.

Step 6: Union bound. Combining Steps 2–5 via the union bound (each of the 4 tail events for σ, μ, $C_{\mathcal{F}_{\text{ro}}}$, and calibration), we obtain:

$$\mathbb{P}(|\widehat{S} - S^*| > \varepsilon) \leq 4\exp\left(-\frac{L\varepsilon^2}{8}\right) + (n_{\text{cal}}^{-1/2}), \tag{42}$$

as stated. □

Corollary 5.6 (Cercis Score as Audit Quality Certificate). *For a target certification confidence $1 - \gamma$, the* Cercis *score $\widehat{S}(\mathbf{s})$ certifies that the true quality $S^*(\mathbf{s})$ lies within $\widehat{S}(\mathbf{s}) \pm \varepsilon$ with:*

$$\varepsilon = \sqrt{\frac{8}{L}\log\frac{4}{\gamma}} + (n_{cal}^{-1/2}). \tag{43}$$

For $L = 10$ and $\gamma = 0.05$, this yields $\varepsilon \approx 1.35$ in absolute score units —indicating that certification at high precision requires larger effective laboratory counts or protocol diversity.

Remark 5.7. *Theorem 2 establishes that the* Cercis *score is not merely a heuristic but a statistically consistent estimator of a well-defined true quality metric. The convergence rate depends on L (not merely L), emphasizing that correlated laboratories provide diminishing certification returns —a finding with direct policy implications for NV center quality standardization efforts.*

6 Theorem 3: T_2 Degradation Source Unidentifiability T_2 退化源不可辨识性

Theorem 6.1 (T_2 Degradation Source Unidentifiability 相干时间退化源不可辨识性)**.** ■ ***[Full Proof]***

Let two laboratories ℓ_A, ℓ_B measure the coherence time of NV center samples in states $\mathbf{s}_A$ and $\mathbf{s}_B$, respectively. Suppose the laboratories report $\widehat{T}_2^{(\ell_A)}(\mathbf{s}_A) \neq \widehat{T}_2^{(\ell_B)}(\mathbf{s}_B)$, i.e., a discrepancy in measured coherence times.

Consider three possible causal sources of the discrepancy:

(E1) E_1*: **Surface noise dominance** —one sample has higher surface spin noise due to different surface termination or shallower NV implantation depth, causing genuine T_2 degradation;*

(E2) E_2*: **Bulk impurity dominance** —one sample has higher bulk paramagnetic impurity concentration (P1 centers, ^{13}C nuclear spins), causing genuine T_2 degradation;*

(E3) E_3*: **Measurement protocol artifact** —the discrepancy arises not from sample differences but from the laboratories using different measurement protocols with distinct spectral filter functions, sensitivity to environmental noise, or calibration procedures.*

Then, without at least one of the following declared structural assumptions —

(S1) Known depth profile of NV centers in both samples, including implantation energy, annealing conditions, and surface characterization (XPS, SIMS);

(S2) Known bulk impurity concentrations (EPR for P1 centers, SIMS for nitrogen, Raman for ^{13}C isotopic abundance) for both samples;

(S3) Declared measurement protocol specifications: pulse sequence, filter function $F_p(\omega)$, microwave power calibration, laser repumping scheme, and environmental noise spectrum $S(\omega)$ at each laboratory —

*the true causal source among $\{E_1, E_2, E_3\}$ is **fundamentally unidentifiable** from the observed T_2 discrepancy alone.*

Furthermore, for any desired posterior distribution $\mathbf{q} = (q_1, q_2, q_3)$ over the three sources (with $q_i \geq 0$, $\sum_i q_i = 1$), there exists a consistent pair of measurement configurations $(\ell_A, \mathbf{s}_A, p_A)$ and $(\ell_B, \mathbf{s}_B, p_B)$ that produces the observed T_2 discrepancy while having exactly that posterior over causal sources.

Proof. We construct three observationally equivalent worlds, each attributing the same T_2 discrepancy to a different dominant cause. This is a proof by construction of non-identifiability in the latent variable model.

Step 1: Latent variable specification. Define three binary indicator variables:

$$Z_1 = \mathbf{1}[T_2 \text{ discrepancy dominated by surface noise } (E_1)], \tag{44}$$

$$Z_2 = \mathbf{1}[T_2 \text{ discrepancy dominated by bulk impurities } (E_2)], \tag{45}$$

$$Z_3 = \mathbf{1}[T_2 \text{ discrepancy dominated by protocol artifact } (E_3)]. \tag{46}$$

These are mutually exclusive: $Z_1 + Z_2 + Z_3 = 1$.

Step 2: Observation model. The observed data is the ordered pair of measured coherence times:

$$\mathbf{o} = (\widehat{T}_2^{(\ell_A)}(\mathbf{s}_A), \widehat{T}_2^{(\ell_B)}(\mathbf{s}_B)) \in \mathbb{R}_+^2, \tag{47}$$

with $\widehat{T}_2^{(\ell_A)} \neq \widehat{T}_2^{(\ell_B)}$. Define the discrepancy magnitude $\delta = |\widehat{T}_2^{(\ell_A)} - \widehat{T}_2^{(\ell_B)}| > 0$.

The mapping from latent causes to observed measurements is parameterized by:

$$\theta_i = (T_2(\mathbf{s}_A; Z_i = 1), T_2(\mathbf{s}_B; Z_i = 1), \beta_{\ell_A}(p_A, \mathbf{s}_A; Z_i = 1), \beta_{\ell_B}(p_B, \mathbf{s}_B; Z_i = 1)), \quad i \in \{1, 2, 3\}, \tag{48}$$

where β are systematic biases. The observation likelihood given cause $Z_i = 1$ is:

$$\mathcal{L}_i = \prod_{m \in \{A,B\}} \phi\left(\frac{\widehat{T}_2^{(\ell_m)} - T_2(\mathbf{s}_m; Z_i = 1) - \beta_{\ell_m}}{\sigma_{\ell_m}}\right), \tag{49}$$

where ϕ is the standard normal density.

Step 3: World 1 —Surface noise dominates ($Z_1 = 1$). Choose $\mathbf{s}_A$ to be a shallow NV center (depth $d_A = 5$ nm) with hydrogen-terminated surface, and $\mathbf{s}_B$ to be a deep NV center (depth $d_B = 50$ nm) in the same diamond substrate. The surface noise spectral density $S_{\text{surf}}(\omega) \propto 1/d^2$ makes $T_2(\mathbf{s}_A) \ll T_2(\mathbf{s}_B)$. Both laboratories use Hahn echo ($p_A = p_B =$ Hahn) with identical apparatus ($\beta_A = \beta_B = 0$). The observed discrepancy δ equals $T_2(\mathbf{s}_B) - T_2(\mathbf{s}_A)$, purely from surface noise.

With parameter choices $T_2(\mathbf{s}_A; Z_1{=}1) = 100\ \mu$s, $T_2(\mathbf{s}_B; Z_1{=}1) = 500\ \mu$s, identical protocols and calibration, the likelihood is:

$$\mathcal{L}_1 = \phi(0) \cdot \phi(0) = \frac{1}{2\pi\sigma^2}. \tag{50}$$

Step 4: World 2 —Bulk impurities dominate ($Z_2 = 1$). Choose $\mathbf{s}_A$ to be an NV center in HPHT diamond with [N] = 100 ppm, and $\mathbf{s}_B$ to be an NV center in electronic-grade CVD diamond with [N] < 1 ppb. Both centers are at the same depth ($d_A = d_B = 50$ nm, eliminating surface noise), both laboratories use Hahn echo. The paramagnetic spin bath density in $\mathbf{s}_A$ causes $T_2(\mathbf{s}_A) \ll T_2(\mathbf{s}_B)$.

With parameter choices $T_2(\mathbf{s}_A; Z_2{=}1) = 100\ \mu$s (identical value as World 1), $T_2(\mathbf{s}_B; Z_2{=}1) =$

500 μs (identical value as World 1), the likelihood is:

$$\mathcal{L}_2 = \phi(0) \cdot \phi(0) = \frac{1}{2\pi\sigma^2} = \mathcal{L}_1. \tag{51}$$

Step 5: World 3 —Protocol artifact dominates ($Z_3 = 1$). Choose $\mathbf{s}_A = \mathbf{s}_B$ (identical NV center, eliminating both surface and bulk heterogeneity). Let laboratory A use Hahn echo while laboratory B uses XY8 dynamical decoupling. The true T_2 is identical for both samples, but the different filter functions $F_{\text{Hahn}}(\omega)$ and $F_{\text{XY8}}(\omega)$ produce different measured values: $\widehat{T}_2^{(\ell_A)} = T_{2\text{true}} + \delta_{\text{Hahn}}$, $\widehat{T}_2^{(\ell_B)} = T_{2\text{true}} + \delta_{\text{XY8}}$, with $\delta_{\text{XY8}} > \delta_{\text{Hahn}}$.

With parameter choices $T_2(\mathbf{s}_A; Z_3{=}1) = T_2(\mathbf{s}_B; Z_3{=}1) = 300$ μs, $\beta_A = -200$ μs (Hahn echo systematic underestimation relative to true T_2), $\beta_B = +200$ μs (XY8 systematic overestimation), the observed measurements are $\widehat{T}_2^{(\ell_A)} = 100$ μs and $\widehat{T}_2^{(\ell_B)} = 500$ μs — identical to Worlds 1 and 2. The likelihood is:

$$\mathcal{L}_3 = \phi(0) \cdot \phi(0) = \frac{1}{2\pi\sigma^2} = \mathcal{L}_1 = \mathcal{L}_2. \tag{52}$$

Step 6: General impossibility of causal attribution. The three worlds produce identical observations $\mathbf{o} = (100\ \mu\text{s}, 500\ \mu\text{s})$ with identical likelihoods $\mathcal{L}_1 = \mathcal{L}_2 = \mathcal{L}_3$, yet each attributes the discrepancy to a fundamentally different physical cause. The parameter space has 12 free parameters (4 parameters per cause $\times$ 3 causes), constrained by one observed discrepancy δ. The solution manifold has dimension $12 - 1 - 2 = 9$ (accounting for the measurement pair and the probability simplex constraints), meaning there is a 9-dimensional family of configurations that produce exactly the same observations.

For any desired posterior $\mathbf{q}^*$, we can interpolate between the three extremal worlds to achieve it: set $T_2(\mathbf{s}_A) = q_1^* \cdot 100 + q_2^* \cdot 100 + q_3^* \cdot 300$, $T_2(\mathbf{s}_B) = q_1^* \cdot 500 + q_2^* \cdot 500 + q_3^* \cdot 300$, and similarly for biases, preserving the observed measurements while realizing the specified posterior.

Step 7: Necessary conditions for identifiability. Each structural assumption (S1)–(S3) breaks the observational equivalence:

- (S1): Known depth profiles distinguish World 1 (different depths) from Worlds 2 and 3 (same depth);
- (S2): Known impurity concentrations distinguish World 2 (different [N]) from Worlds 1 and 3;
- (S3): Declared protocols distinguish World 3 (different protocols) from Worlds 1 and 2 (same protocol).

Without at least one, the three worlds remain indistinguishable. Without all three, at best pairs of causes can be conflated. □

Corollary 6.2 (Practical Implication for NV Center Publication Standards)**.** *Theorem 3 implies that any publication reporting an NV center T_2 value must, at minimum, declare:*

(i) The NV implantation depth (or a measured depth profile);

(ii) The bulk nitrogen and ^{13}C concentrations of the diamond substrate;

(iii) The complete measurement protocol specification, including pulse sequence, repetition rate, laser power, and environmental noise characterization.

*Without all three declarations, the reported T_2 value cannot be causally interpreted —it could reflect genuine material quality, or it could be an artifact of measurement choices. This is a minimum information standard (**NV-MIS**) for NV center characterization.*

Remark 6.3 (Relation to Derman's Principle of Model Risk)**.** *Theorem 3 embodies, in the quantum materials context, Derman's framework of model risk [**?**]: the source of a discrepancy between measurement and expectation (here, between two laboratories' T_2 measurements) cannot be resolved without declared assumptions about the generative process. Our construction of three observationally equivalent worlds formalizes this principle for NV center auditing.*

7 Cercis Score and Yajie Multi-Laboratory Consensus Cercis 评分与 Yajie 多实验室共识

7.1 Cercis Score: Detailed Formulation

The Cercis score $S(\mathbf{s}) = Q(\mathbf{s}) + \eta N(\mathbf{s})$ was defined in Definition **??**. We now provide the fully operationalized form for NV center quality auditing.

Definition 7.1 (Operationalized Quality Concordance 可操作的品质一致性)**.** *For state* $\mathbf{s}$ *audited by laboratories $\mathcal{L}_{audit}$:*

$$Q(\mathbf{s}) = \alpha_T \cdot R_{T_2}(\mathbf{s}) + \alpha_F \cdot C_{\mathcal{F}_{ro}}(\mathbf{s}) + \alpha_\Omega \cdot R_{\Omega_R}(\mathbf{s}), \tag{53}$$

with $\alpha_T + \alpha_F + \alpha_\Omega = 1$, where:

- *$R_{T_2}(\mathbf{s})$ is the T_2 reproducibility (Definition **??**);*
- *$C_{\mathcal{F}_{ro}}(\mathbf{s})$ is the readout fidelity consistency (Definition **??**);*
- *$R_{\Omega_R}(\mathbf{s})$ is the Rabi frequency reproducibility, defined analogously to R_{T_2};*
- *Default weights: $\alpha_T = 0.5, \alpha_F = 0.3, \alpha_\Omega = 0.2$, reflecting the primacy of coherence time as the most information-rich NV quality parameter.*

Definition 7.2 (Operationalized Fabrication Novelty 可操作的制备新颖性)**.** *The fabrication novelty $N(\mathbf{s})$ decomposes across six dimensions, each weighted by its impact on NV center performance:*

$$N(\mathbf{s}) = \sum_{k=1}^{6} v_k \cdot n_k(\mathbf{s}), \tag{54}$$

where:

Dimension	***Weight*** v_k	***Novelty indicator*** $n_k(\mathbf{s})$
Substrate type (CVD/HPHT)	*0.20*	$\mathbf{1}[s_{sub} \notin \mathcal{S}_{cal,sub}]$
Nitrogen isotope (N-14/N-15)	*0.15*	$\mathbf{1}[s_{iso} \notin \mathcal{S}_{cal,iso}]$
Implantation depth regime	*0.25*	$\mathbf{1}[s_{depth} \notin \mathcal{S}_{cal,depth}]$
Surface termination chemistry	*0.15*	$\mathbf{1}[s_{surf} \notin \mathcal{S}_{cal,surf}]$
^{13}*C isotopic enrichment*	*0.15*	$\mathbf{1}[s_{C13} \notin \mathcal{S}_{cal,C13}]$
Annealing/fabrication protocol	*0.10*	$\mathbf{1}[protocol \notin \mathcal{S}_{cal,fab}]$

Proposition 7.3 (Monotonicity of CERCIS Score). *Under Assumptions* **??**–**??**, *the* CERCIS *score satisfies:*

(i) $S(\mathbf{s}) \in [0, 1+\eta]$ *for all* $\mathbf{s} \in \mathcal{S}$*;*

(ii) S *is non-decreasing in each laboratory's measurement precision;*

(iii) $S(\mathbf{s}) \to 1 + \eta N(\mathbf{s})$ *as* $L \to \infty$ *and all laboratories achieve their asymptotic measurement limits.*

Proof. (i) Follows from $R_{T_2}, C_{\mathcal{F}_{\mathrm{ro}}}, R_{\Omega_R} \in [0,1]$ and the convex combination weights summing to 1, with $N(\mathbf{s}) \in [0,1]$.

(ii) Measurement precision enters through $\sigma_{\ell,p}$ in the measurement model (**??**). As precision improves ($\sigma_{\ell,p} \to 0$), the inter-laboratory variance decreases, increasing R_Q monotonically.

(iii) As $L \to \infty$, Theorem 2 implies $\widehat{S} \to S^*$, and with perfect precision $S^* = 1 + \eta N(\mathbf{s})$ (since $\sigma_Q^{\mathrm{true}} \to 0$ and $\delta_{\mathcal{F}_{\mathrm{ro}}}^{\mathrm{true}} \to 0$). □

7.2 Yajie Multi-Laboratory Consensus

Definition 7.4 (YAJIE Multi-Laboratory Consensus 多实验室共识). *Given* L *laboratories with measurements* $\{\widehat{Q}^{(\ell,p_\ell^*)}(\mathbf{s})\}_{\ell=1}^{L}$ *for quality parameter* Q *on state* $\mathbf{s}$*, the* ***Yajie consensus*** *estimate is:*

$$\widehat{Q}_{\text{YAJIE}}(\mathbf{s}) = \sum_{\ell=1}^{L} w_\ell(\mathbf{s}) \cdot \widehat{Q}^{(\ell,p_\ell^*)}(\mathbf{s}), \tag{55}$$

where the weights are:

$$w_\ell(\mathbf{s}) = \frac{S(\mathbf{s}) \cdot \omega_\ell + (1 - S(\mathbf{s})) \cdot \bar{\omega}}{\sum_{\ell'=1}^{L} \big(S(\mathbf{s}) \cdot \omega_{\ell'} + (1 - S(\mathbf{s})) \cdot \bar{\omega}\big)}, \tag{56}$$

with:

- ω_ℓ*: base weight for laboratory* ℓ*, computed as the inverse of its mean squared error on the calibration set* $\mathcal{S}_{cal}$*:* $\omega_\ell = \left(\sum_{\mathbf{s} \in \mathcal{S}_{cal}} (\widehat{Q}^{(\ell,p_\ell^*)}(\mathbf{s}) - Q_{ref}(\mathbf{s}))^2\right)^{-1}$*;*
- $\bar{\omega} = \frac{1}{L}\sum_{\ell=1}^{L} \omega_\ell$*: uniform baseline weight.*

Weighting principle*: When the* CERCIS *score* $S(\mathbf{s})$ *is high (high reproducibility, low novelty —the sample is "well-understood"), weights concentrate on precise laboratories.*

When $S(\mathbf{s})$ is low (poor reproducibility or high novelty —the sample is "unfamiliar"), weights revert to uniform, preventing overconfident extrapolation.

Proposition 7.5 (YAJIE Consensus Optimality Property)**.** *Under Assumptions* **??**–**??***, the* YAJIE *consensus estimator minimizes the expected squared error among all linear estimators with weights in the convex hull of $\{\omega_\ell\}$ and $\bar{\omega}$:*

$$\mathbb{E}[(\widehat{Q}_{\text{YAJIE}} - Q(\mathbf{s}))^2] \leq \min_{\mathbf{w} \in conv(\boldsymbol{\omega}, \bar{\omega}\mathbf{1})} \mathbb{E}[(\widehat{Q}_{\mathbf{w}} - Q(\mathbf{s}))^2] + (L^{-1}). \tag{57}$$

Proof. The YAJIE weights (**??**) are a convex combination of the precision-optimal weights $\boldsymbol{\omega}$ (minimizing variance when biases are zero) and the uniform weights $\bar{\omega}\mathbf{1}$ (minimizing worst-case bias when biases are unknown). By the fundamental bias-variance decomposition, this achieves near-optimal risk for the minimax problem over the uncertainty class defined by $S(\mathbf{s})$, with excess risk (L^{-1}) from finite-sample weight estimation. □

8 Experimental Benchmarks 实验基准

The experimental framework evaluates the SCX auditing theorems across four benchmark axes, each probing a distinct dimension of NV center heterogeneity.

8.1 Benchmark Design

B1: N-14 vs N-15 Isotopic Composition (N-14 与 N-15 同位素对比).

- *Motivation*: N-14 has nuclear spin $I = 1$ (three-level hyperfine structure), while N-15 has $I = 1/2$ (two-level). The hyperfine coupling strength differs ($A_{\text{N-14}} \approx$ 2.16 MHz vs $A_{\text{N-15}} \approx 3.03$ MHz), affecting decoherence channels and readout fidelity.
- *Protocol*: Prepare matched pairs of NV centers —one ensemble with natural N-14, one with isotopically enriched N-15 —in identical CVD diamond substrates at matching depths. Distribute to $L \geq 5$ laboratories for blind measurement of T_2, T_2^*, Ω_R, and $\mathcal{F}_{\text{ro}}$.
- *SCX Metrics*: Compute R_{T_2} for N-14 and N-15 cohorts separately. Test whether the YAJIE consensus detects systematic differences in T_2 that are attributable to isotopic composition rather than measurement noise (Theorem 1 bound application).

B2: Shallow vs Deep NV Centers (浅层与深层 NV 中心对比).

- *Motivation*: Shallow NV centers ($d < 10$ nm) are essential for nanoscale NMR sensing but suffer from surface-induced decoherence. Deep NV centers ($d >$ 50 nm) are bulk-noise-limited. The surface-to-bulk noise transition tests Theorem 3 unidentifiability.

- *Protocol*: Implant NV centers at 5 depths ($d \in \{3, 8, 15, 30, 80\}$ nm) in the same CVD diamond substrate with identical surface termination. Each laboratory measures all depths using Hahn echo. Additionally, each laboratory characterizes surface noise via T_1 relaxometry at variable depth (structural assumption S1 satisfaction).
- *SCX Metrics*: Fit $1/T_2(d) = 1/T_2^{\text{bulk}} + \gamma/d^2$ (surface noise model). Measure the inter-laboratory variance of the fitted T_2^{bulk} parameter. Low variance certifies that surface noise is correctly accounted for (validating S1).

B3: CVD vs HPHT Diamond Substrates (CVD 与 HPHT 金刚石衬底对比).

- *Motivation*: CVD and HPHT diamond differ systematically in nitrogen incorporation, strain distribution, and defect density. CVD diamond can achieve [N] $<$ 1 ppb, while HPHT diamond typically has [N] $\sim$ 1–200 ppm. These differences directly impact T_2.
- *Protocol*: Procure matched pairs: (i) electronic-grade CVD ([N] $<$ 5 ppb, ^{13}C $<$ 0.01%), (ii) standard CVD ([N] $\sim$ 10–50 ppb), (iii) type-IIa HPHT ([N] $<$ 1 ppm), (iv) type-Ib HPHT ([N] $\sim$ 100 ppm). All with deep NV centers ($d =$ 50 nm) to eliminate surface effects. Audit by $L \geq 5$ laboratories.
- *SCX Metrics*: CERCIS score $S(\mathbf{s})$ for each substrate type. Q component should be highest for electronic-grade CVD (best reproducibility), while N component captures the novelty of isotopically engineered substrates.

B4: Cross-Protocol Measurement Comparison (跨协议测量对比).

- *Motivation*: Different measurement protocols probe different noise spectral regions (Assumption **??**). The relationship between T_2^* (ODMR), T_2 (Hahn echo), and $T_{2,\text{DD}}$ (XY8) provides information about the noise spectrum $S(\omega)$.
- *Protocol*: For a single NV center sample (state fixed), each laboratory performs all four protocols: ODMR (T_2^*), Ramsey (T_2^* with different filter), Hahn echo (T_2), and XY8 ($T_{2,\text{XY8}}$). Report all four values with measurement uncertainties.
- *SCX Metrics*: The protocol amplification ratio $\mathcal{A} = T_{2,\text{XY8}}/T_2^*$ quantifies the noise spectrum slope. Inter-laboratory consistency of $\mathcal{A}$ is a protocol-independent quality metric. If $\mathcal{A}$ varies across laboratories on the same sample, this indicates unaccounted systematic biases (Theorem 1 detection). The unidentifiability of whether $\mathcal{A}$ variation arises from protocol implementation differences or genuine environmental noise differences is established by Theorem 3.

8.2 Evaluation Metrics

M1: Consensus Error Rate. Compare $\mathbb{P}(|\widehat{Q}_{\text{YAJIE}} - Q_{\text{ref}}| > \delta)$ against the Theorem 1 bound $\exp(-2L\Delta^2)$. For calibration samples with known reference values Q_{ref}, measure the empirical tail probability and test whether it respects the Hoeffding bound.

M2: Effective Laboratory Count L. Estimate $\bar{\rho}$ from the empirical error correlation matrix on calibration samples and compute $L = L/(1 + (L-1)\bar{\rho})$. Measure how L

degrades when laboratories share equipment vendors, calibration standards, or data analysis software.

M3: Cercis Score Calibration. For samples with known fabrication parameters and independently verified quality (e.g., by multi-lab consensus on calibration set), compute $\Gamma(S) = \mathbb{E}[|\mathbb{E}[Q_{\text{ref}} \mid S] - S|]$. A well-calibrated CERCIS score has $\Gamma(S) \approx 0$.

M4: Unidentifiability Verification. For the depth-series benchmark (B2), test whether laboratories can distinguish surface-dominated from bulk-dominated decoherence using only T_2 measurements (without S1 depth information). The prediction from Theorem 3 is that they cannot.

M5: Novelty Detection. For diamond samples fabricated by novel methods (new surface treatments, exotic substrates, novel implantation techniques), measure the CERCIS novelty component $N(\mathbf{s})$. High N should correlate with high inter-laboratory measurement variance, validating N as a predictor of certification difficulty.

8.3 Ablation Study

To empirically validate the theorems, we specify the following ablation:

A1: Full audit cohort ($L = 10$). All laboratories active, full protocol diversity. Measure L and consensus error rate. Compare against Theorem 1 bound.

A2: Remove correlated laboratories. Identify laboratory pairs with $\rho_{\ell\ell'} > 0.5$ (shared equipment, shared calibration). Drop one from each pair. Expected: L decreases by less than ΔL, demonstrating the sub-additivity of correlated information.

A3: Reduce protocol diversity ($\mathcal{P}_\ell = \{\textbf{Hahn}\}$ for all ℓ). All laboratories use only Hahn echo. Expected: L drops sharply because protocol correlation $\kappa_{pp'} = 1$ for identical protocols, inflating $\bar{\rho}$.

A4: Vary calibration set size. Use $n_{\text{cal}} \in \{5, 10, 20, 40\}$. Verify that the $(1/\sqrt{n_{\text{cal}}})$ calibration error term in Theorem 2 scales as predicted.

A5: Test unidentifiability claim. Provide laboratories with only T_2 measurements from B2 (depth series) without depth information. Ask them to classify each sample as "surface-limited" or "bulk-limited." Theorem 3 predicts classification accuracy at chance level (33% for 3-class, 50% for 2-class), confirming unidentifiability without S1.

9 Discussion 讨论

9.1 Summary of Contributions

This paper establishes the mathematical foundation for auditable quality certification of nitrogen-vacancy centers in diamond. The three theorems address complementary aspects of the audit problem:

1. **Theorem 1** provides an exponential error detection guarantee: the probability that L effectively independent laboratories all miss a systematic measurement error decays as $\exp(-2L\Delta^2)$. This formalizes the intuition that multi-laboratory auditing provides robustness, while the L correction (accounting for inter-lab correlation) prevents overconfidence from nominally large but correlated laboratory cohorts.
2. **Theorem 2** establishes that the CERCIS score is a statistically consistent estimator of true NV center quality, converging at rate $(\exp(-L\varepsilon^2/8))$ in the effective laboratory count. This legitimizes the CERCIS score as more than a heuristic —it is a proper scoring rule for quantum device quality.
3. **Theorem 3** proves that the causal attribution of T_2 degradation —whether from surface noise, bulk impurities, or measurement protocol artifacts —is fundamentally unidentifiable without declared structural assumptions. This is the NV center analogue of Derman's model risk principle, with direct implications for publication standards.

9.2 The NV-MIS Publication Standard

As a direct consequence of Theorem 3, we propose the **NV Minimum Information Standard (NV-MIS)** for any publication reporting NV center coherence properties. Every paper claiming an NV center T_2, T_1, or $\mathcal{F}_{\mathrm{ro}}$ value must declare:

MIS-1: **Depth**: NV implantation energy, simulated or measured depth distribution (mean $\pm$ straggle), and depth verification method (e.g., NMR depth calibration, ion channeling);

MIS-2: **Impurities**: Bulk nitrogen concentration (EPR or SIMS), ^{13}C isotopic abundance (SIMS or Raman), and other identified paramagnetic defect concentrations;

MIS-3: **Protocol**: Complete pulse sequence specification, microwave and laser parameters, repetition rate, data acquisition duration, environmental conditions (temperature, magnetic field stability);

MIS-4: **Surface**: Surface termination chemistry, cleaning procedure, and any post-implantation treatment (annealing temperature, duration, atmosphere);

MIS-5: **Substrate**: Diamond growth method (CVD/HPHT), vendor, substrate orientation ($\{100\}$, $\{111\}$, $\{110\}$), and any pre-treatment.

Journals adopting NV-MIS would eliminate the audit gap identified in Section **??**: a reviewer could, in principle, determine whether a reported T_2 value is consistent with the declared physical parameters or whether it represents an extraordinary claim requiring extraordinary multi-laboratory verification.

9.3 Limitations and Honest Caveats

We state these limitations explicitly:

L1: Gaussian noise assumption. The measurement model (**??**) assumes Gaussian noise. Non-Gaussian noise (e.g., telegraph noise from fluctuating charges, $1/f$ noise in the electronics) would require heavier-tailed concentration inequalities (e.g., Bernstein bounds).

L2: Stationary noise spectra. The analysis assumes time-stationary noise spectra $S(\omega)$. Real NV centers experience spectral diffusion (slow fluctuations in the noise environment) that violates stationarity on timescales of minutes to hours. The theorems' guarantees degrade under non-stationarity by a factor proportional to the spectral diffusion rate.

L3: Finite calibration set. The $(1/\sqrt{n_{\text{cal}}})$ term in Theorem 2 becomes non-negligible for small n_{cal}. A calibration set of $n_{\text{cal}} \geq 50$ is recommended for certification applications.

L4: Protocol diversity is bounded. The four protocols in $\mathcal{P}$ are not fully independent; all share the same optical readout stage. A genuinely orthogonal measurement (e.g., electrical readout via photocurrent detection) would increase effective information but is not widely deployed.

L5: No quantum supremacy claims. This paper makes no claim about quantum computational advantage. NV centers are a platform for quantum sensing, networking, and information processing; the certification framework is orthogonal to the question of whether any quantum device outperforms classical computation.

9.4 Comparison with Existing Approaches

Current approaches to NV center quality assurance include:

- **Single-lab characterization with error bars**: The dominant paradigm. Provides no cross-validation against systematic biases.
- **Round-robin inter-laboratory comparisons**: Ad-hoc studies (e.g., "Q-NMR Round Robin" efforts) that measure the same sample across labs but lack a theoretical framework for interpreting disagreement.
- **Standard reference materials**: NIST-traceable diamond samples with NV centers of known properties. Valuable but limited to specific parameter regimes and not universally available.

The SCX framework differs from all three by providing (i) formal probabilistic guarantees via Theorems 1–3, (ii) a scalar audit metric (Cercis score) that integrates reproducibility and novelty, and (iii) a weighting mechanism (Yajie consensus) that adapts to calibration uncertainty.

9.5 Future Directions

1. **Extension to other quantum platforms**: The SCX auditing framework applies equally to superconducting qubits (T_2, gate fidelity), trapped ions, and silicon spin qubits. The state space and protocols change, but the mathematical structure is invariant.

2. **Dynamic auditing**: Continuous monitoring of NV center quality over time (aging, radiation damage, surface degradation) within the SCX framework.
3. **Cryptographic certification**: Integrating the SCX audit with blockchain-anchored measurement logs for tamper-proof quality certification.
4. **Automated protocol selection**: Using the SPRING gating mechanism to automatically select optimal measurement protocols based on real-time noise characterization.

10 Conclusion 结论

The nitrogen-vacancy center in diamond is a remarkable quantum system —but its quality claims are only as credible as the audit framework that certifies them. This paper has provided that framework.

We have formalized multi-laboratory NV center characterization as an SCX multi-expert consensus problem, where laboratories are experts, measurement protocols are modalities, and diamond heterogeneity defines the state space. Three theorems with full proof (■ **[Full Proof]**)*establish* :

- Exponentially strong error detection guarantees under multi-laboratory auditing (Theorem 1), with the effective laboratory count L correcting for inter-laboratory correlation from shared resources;
- Statistical consistency of the CERCIS score as a quality metric, converging to true quality at a rate controlled by L and calibration set size (Theorem 2);
- Fundamental unidentifiability of T_2 degradation sources (surface noise vs bulk impurities vs protocol artifacts) without declared structural assumptions (Theorem 3).

The CERCIS score $S = Q + \eta N$ and the YAJIE multi-laboratory consensus provide operational tools for quantum device certification. The NV-MIS publication standard codifies the minimum information required for auditable NV center quality claims.

The central message is that auditability is not an afterthought for quantum device characterization —it is a prerequisite for scientific credibility. A coherence time measured by one laboratory with one protocol on one diamond sample is not a certified property of the NV center; it is a claim that requires multi-laboratory verification under the framework developed here. The mathematics shows that this verification can be exponentially strong, that the CERCIS score provides a principled quality metric, and that without declaring assumptions, the sources of discrepancy are fundamentally unidentifiable.

No quantum supremacy is claimed. The supremacy is in the audit.

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